

PHIGS Workstations & DirectX

- A Journey Through Computer Graphics Evolution

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Agenda

- What We'll Explore

- Historical context of graphics standards
- PHIGS: Architecture, workstation model, and transformations
- Introduction to Microsoft DirectX and its evolution
- DirectX 12 Ultimate: Core technologies (Raytracing, Mesh Shaders, VRS)
- Summary and comparison

Introduction to Graphics Standards

- The Need for Standardization
 - Problem: Early graphics programming was device-specific and non-portable
 - Goal: Create device-independent ways to develop graphical applications
 - Two Approaches:-
 - a. **Formal Standards (ISO)**: e.g., GKS, PHIGS (focused on consistency)
 - b. **Industry Standards**: e.g., OpenGL, DirectX (focused on performance)

What is PHIGS?

- Programmer's Hierarchical Interactive Graphics System
 - ANSI/ISO standard (ISO/IEC 9592) for 3D graphics, prominent in the 1980s-90s
 - Used in professional/scientific workstations (CAD, simulation)
 - Key Innovation: Centralized Structure Store (CSS) - a hierarchical scene graph for managing complex objects
 - PHIGS+: Later extension adding lighting, shading, and depth cueing

The PHIGS Workstation Concept

- An Abstract Graphics Environment
- In PHIGS, a "workstation" is not just a physical device - it's a logical abstraction.
- It represents a complete graphics environment with:-
 - a. A display surface (screen, window, plotter paper)
 - b. Input capabilities (keyboard, mouse, digitizer)
 - c. A workstation state list (current attributes and settings)

The PHIGS Workstation Concept

- Key Types:
 - a. **OUTPUT**: Plotters, film recorders
 - b. **OUTIN**: Interactive terminals (most common)
 - c. **INPUT**: Digitizers
- PEX: PHIGS Extension to X (implemented PHIGS in Unix/X11 environments)

Workstation Configuration

- Opening and Initializing a PHIGS Workstation

To use a workstation, an application calls:-

OPEN WORKSTATION (ws, conid, wstype)

Parameter	Function
ws	Application-defined workstation ID
conid	Connection identifier (file path, network connection)
wstype	Links to a Workstation Description Table

This table tells PHIGS what the device can and cannot do, ensuring portability.

Workstation Transformation

- Mapping the Virtual Scene to the Display

After viewing transforms produce a picture in Normalized Projection Coordinates (NPC) (unit square 0,0 to 1,1):

1. Set Workstation Window: Defines which rectangular region of the NPC unit square to display
2. Set Workstation Viewport : Defines where on the physical display to map that window
 - **Critical**: Aspect ratio of window and viewport must match to avoid distortion (circles become ellipses)
 - **Update Workstation** : Forces a redraw after changes

PHIGS - Legacy

- Strengths:
 - a. Formal standard with strict specification
 - b. Powerful hierarchical data management (influenced later scene graphs)
 - c. True device independence
- Weaknesses:
 - a. Complex and cumbersome to implement efficiently
 - b. Lacked direct hardware control (performance suffered)
 - c. PHIGS+ arrived too late; couldn't keep up with rapid innovation
- Legacy: Superseded by OpenGL in the professional space by the mid-1990s

Introduction to DirectX

- Microsoft's Multimedia Powerhouse
- A collection of low-level APIs for high-performance multimedia on Windows platforms
- Origin: Created to give developers direct hardware access, overcoming GDI's limitations
- Core Components:
 - a. Direct3D (D3D): 3D graphics rendering (our focus)
 - b. Direct2D: 2D graphics
 - c. DirectWrite: Text rendering
 - d. DirectCompute: GPGPU (General-Purpose GPU computing)
 - e. DirectX Diagnostics (Dxdiag): Troubleshooting tool

The Evolution of DirectX

- A Timeline of Innovation

- DirectX 9 (2002) : Long-lived, versatile. Shader Model 2.0/3.0. Still a fallback for compatibility
- DirectX 10 (2006) : Major overhaul, unified shader architecture, reduced CPU overhead
- DirectX 11 (2009) : Added tessellation, multithreaded rendering, compute shaders
- DirectX 12 (2015) : "Close to the metal" API, drastically reducing CPU overhead, finer GPU control
- DirectX 12 Ultimate (2020) : Unified feature set across PC and Xbox Series X/S

DirectX 12 Ultimate

- The New Gold Standard
- Goal: Codify a standard feature set for next-generation graphics across PC and console
- Four Pillar Technologies:
 - a. DirectX Raytracing (DXR) 1.1: Hardware-accelerated real-time ray tracing
 - b. Variable Rate Shading (VRS): Vary shading rate across the frame for performance
 - c. Mesh Shaders: Flexible geometry pipeline replacing vertex / tessellation shaders
 - d. Sampler Feedback: Intelligent texture streaming for massive worlds

DirectX Raytracing (DXR)

- Cinema-Quality Light in Real-Time
- It stimulates physical light behavior for realistic rendering.
- DXR 1.1 (Part of DirectX 12 Ultimate): Adds inline ray tracing for developer flexibility
- Enables Techniques:
 - a. Realistic reflections and shadows
 - b. Global Illumination (multi-bounce indirect lighting)
 - c. RTX Direct Illumination (millions of dynamic lights)
- Challenge: Computationally expensive - requires denoising (e.g., NVIDIA Real-Time Denoiser) to clean up low-sample signals

Mesh Shaders

- Reinventing the Geometry Pipeline
- The Old Way (Vertex/Tessellation Shaders) : Rigid, sequential pipeline with fixed stages
- The Mesh Shader Way: Replaces the entire pipeline with two programmable stages:-
 - a. Task Shader (optional): Coarse culling and amplification
 - b. Mesh Shader: Outputs compact meshlets (groups of primitives) efficiently
- Benefit: Massive flexibility for GPU-driven geometry culling and LOD (Level of Detail), boosting performance in complex scenes

Variable Rate Shading (VRS)

- Shading Smarter, Not Harder
- Problem: Shading every pixel at full rate is wasteful in areas where detail isn't noticeable
- Solution (VRS): Dynamically change the shading rate per region
- Shading Rates:
 - a. 1x1: Full shading (1 shading sample per pixel)
 - b. 2x2: One shading sample for a 2x2 pixel block (lower quality, higher performance)
 - c. Other rates (2x1, 1x2) available
- Use Cases: Apply lower rates to motion-blurred areas, peripheral vision, or dark regions.

Summary & Conclusion

- From Structured Hierarchies to Ultimate Control

Feature	PHIGS	DirectX 12 Ultimate
Paradigm	Retained-mode (scene graph)	Low-level, explicit control
Hardware Access	Abstracted, device-independent	"Close to the metal"

Primary Use	Professional, scientific, CAD	Gaming, high-performance graphics
Key Strengths	Hierarchical data, standardization	Performance, cutting-edge features
Weakness	Performance overhead, rigid	Steep learning curve, platform-specific

Conclusion: PHIGS formalized key concepts like hierarchical structures; DirectX delivers maximum performance and hardware control. The evolution mirrors computing's shift from centralized standards to specialized, high-performance tools.



THANK YOU!!!

Any Queries?